Introduction to Cybersecurity

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Lecture 8

Lecture Rules

Cyberattacks and Vulnerabilities
Part II

Lecture Outline

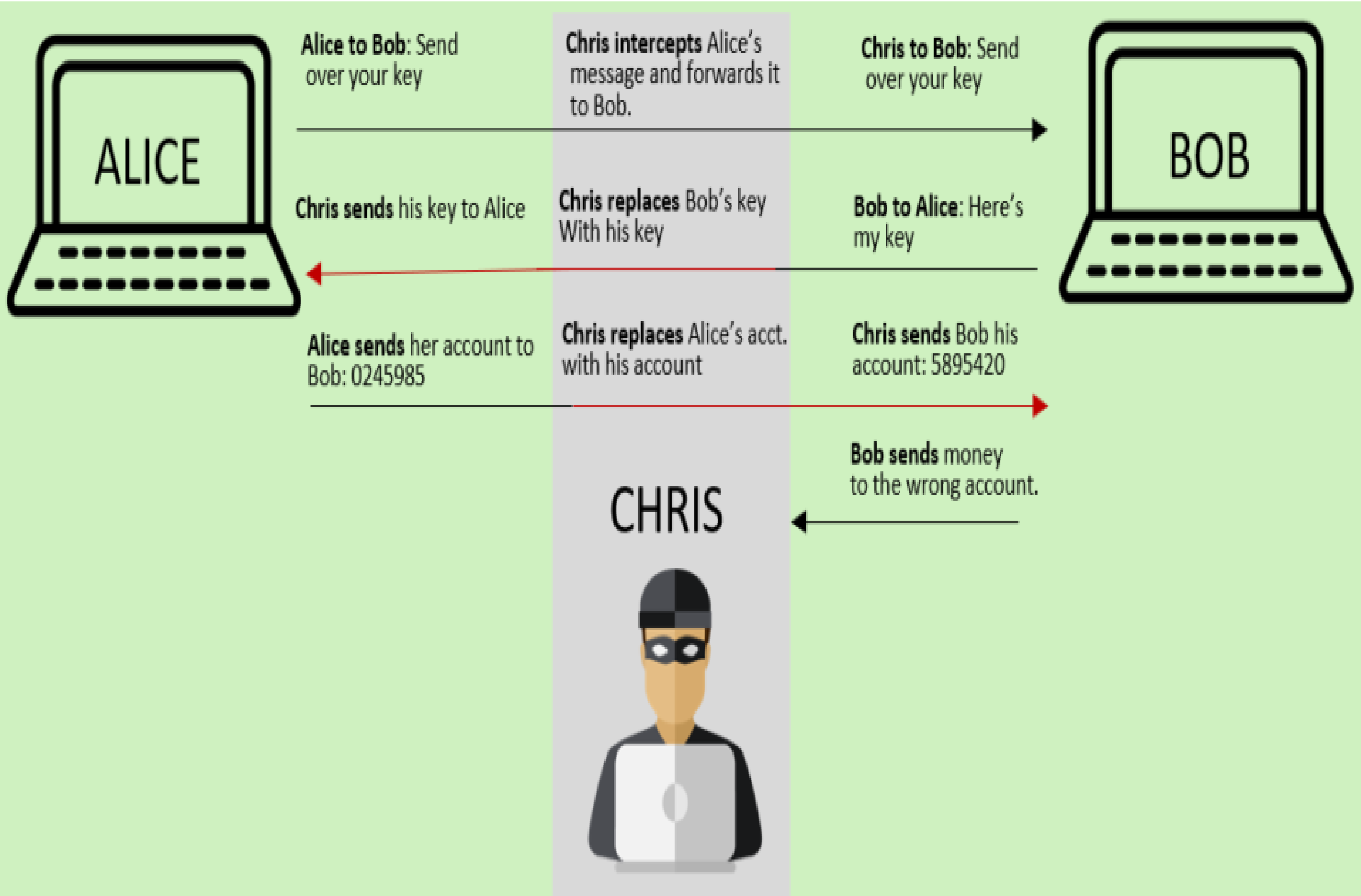
- **Man-In-The-Middle (MITM) attack**
- **SQL Injection attack**
- **Spamming attack**
- **Zero-Day Exploitation attack**
- **Phishing attack**

Man-In-The-Middle (MITM) Attack

- The **hacker intercepts** the **normal connection** between the **user** and the **web server** without any **knowledge** of both **user** and **server**.

- The **legitimate communication** link between the two **entities** is exploited, intercepted, and **decrypted** to **steal** the **personal information** for **malicious** use.

Man-In-The-Middle (MITM) Attack

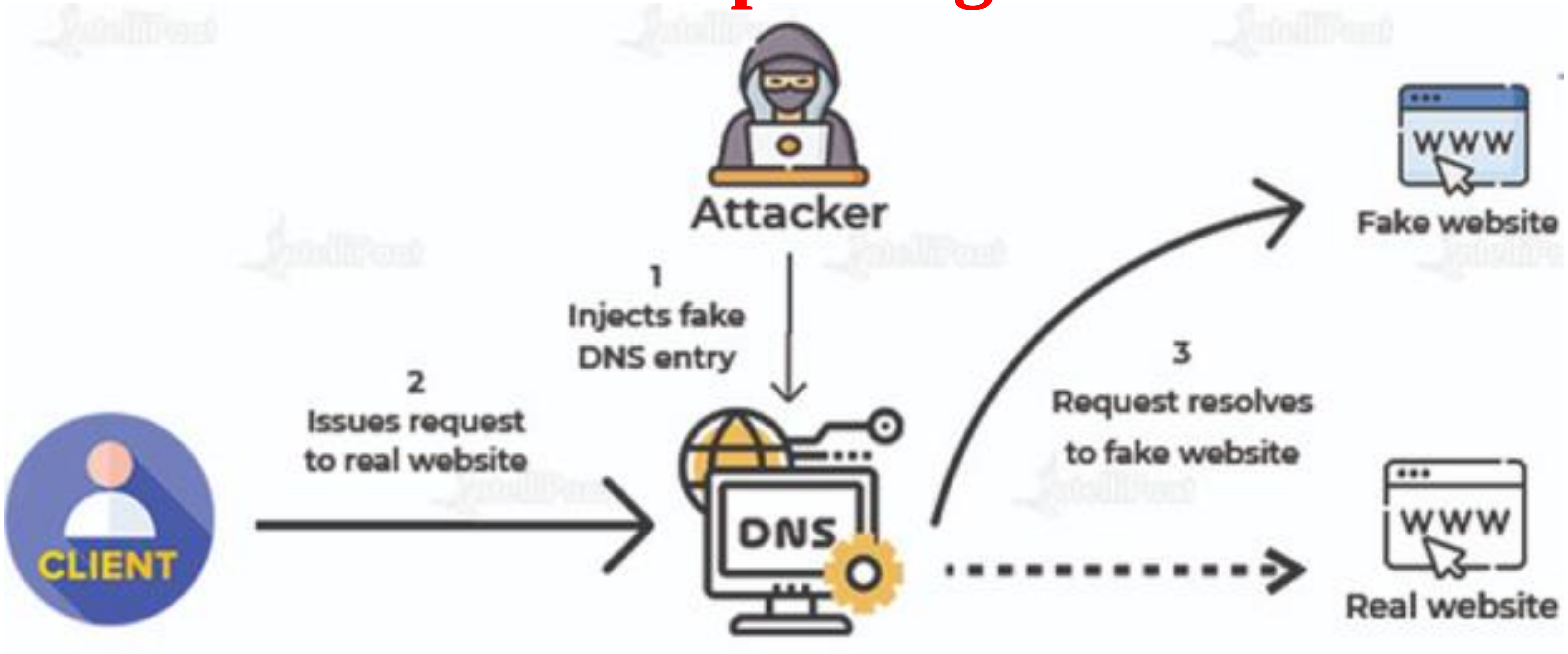


3.Man-In-The-Middle (MITM) Attacks

- **The major types of MITM attacks include the following.**
  - **DNS spoofing.**
  - **HTTP spoofing.**
  - **IP spoofing.**
  - **Email hijacking.**
  - **SSL (Secure Sockets Layer) hijacking.**
  - **Wi-Fi network eavesdropping.**
  - **Stealing the cookies set on the browsers.**

Man-In-The-Middle (MITM) Attack

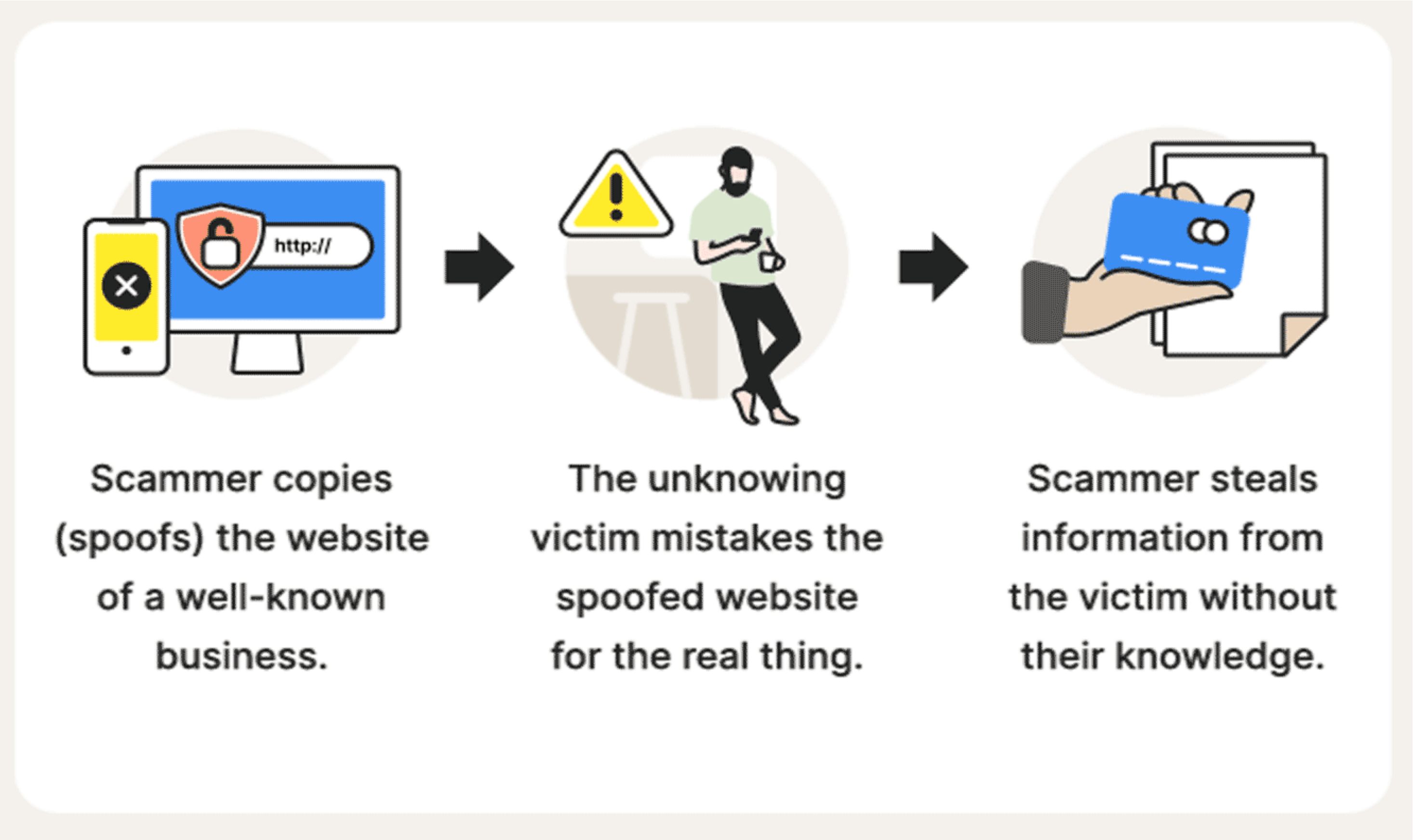
DNS spoofing



- When **hackers** trick your computer into going to a **fake website** instead of the **real** one by **manipulating** the internet's "phonebook" **(DNS).**
- DNS **translates human-readable** domain names (**like google.com**) into **numerical** IP addresses **(like 172.217.160.142)** that **computers** use to **communicate** with each other.

Man-In-The-Middle (MITM) Attack

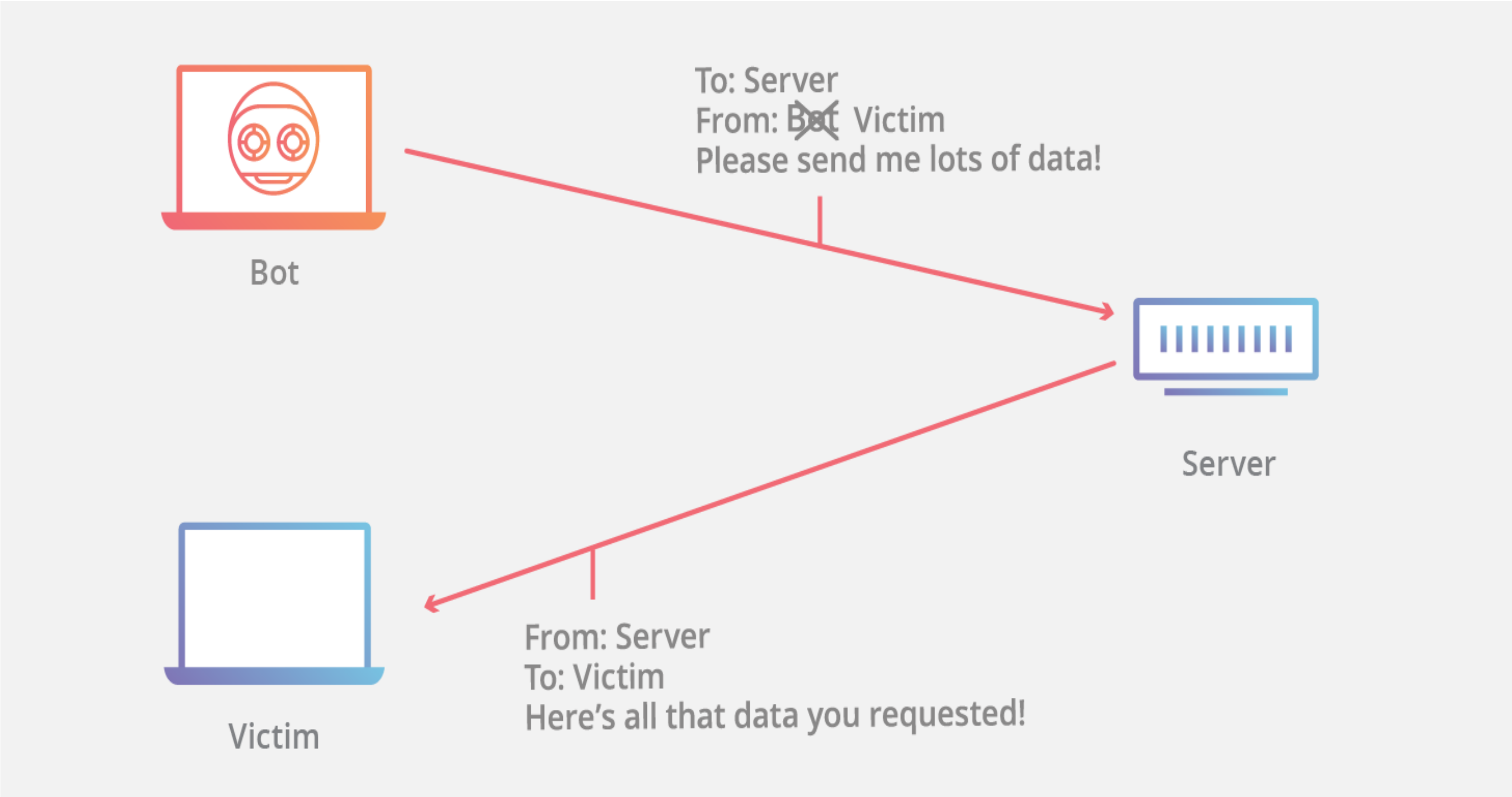
HTTP spoofing



- When **hacker** creates a **fake copy** of a **real website** to **trick users** into giving up **personal information**.

Man-In-The-Middle (MITM) Attack

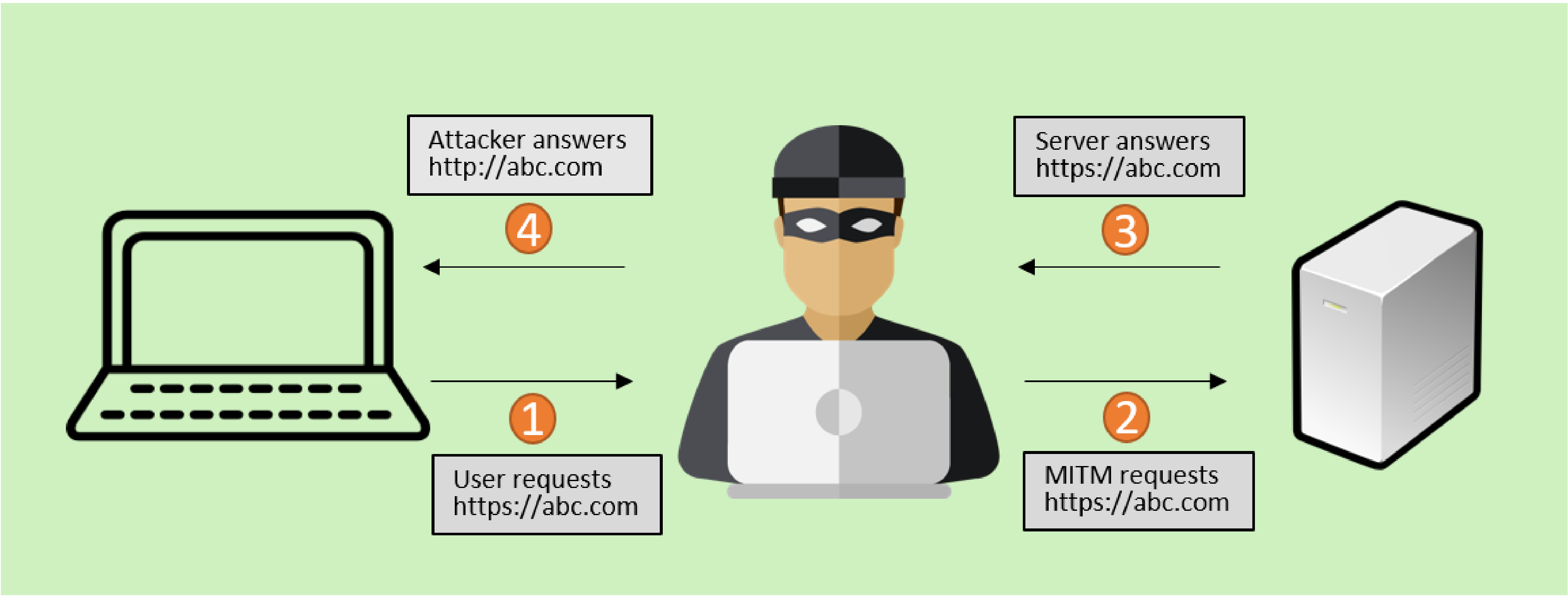
IP spoofing



- An attacker **manipulates** the **source IP address** in network **packets** to make it **appear** that the **packets** come from a **trusted source**.

Man-In-The-Middle (MITM) Attack

SSL Hacking



- Attackers **exploit** **weaknesses** in the **Secure Sockets Layer (SSL)** or **Transport Layer Security** **(TLS),** to **intercept**, **decrypt**, or **manipulate encrypted communication** between a **user** and a **website**.

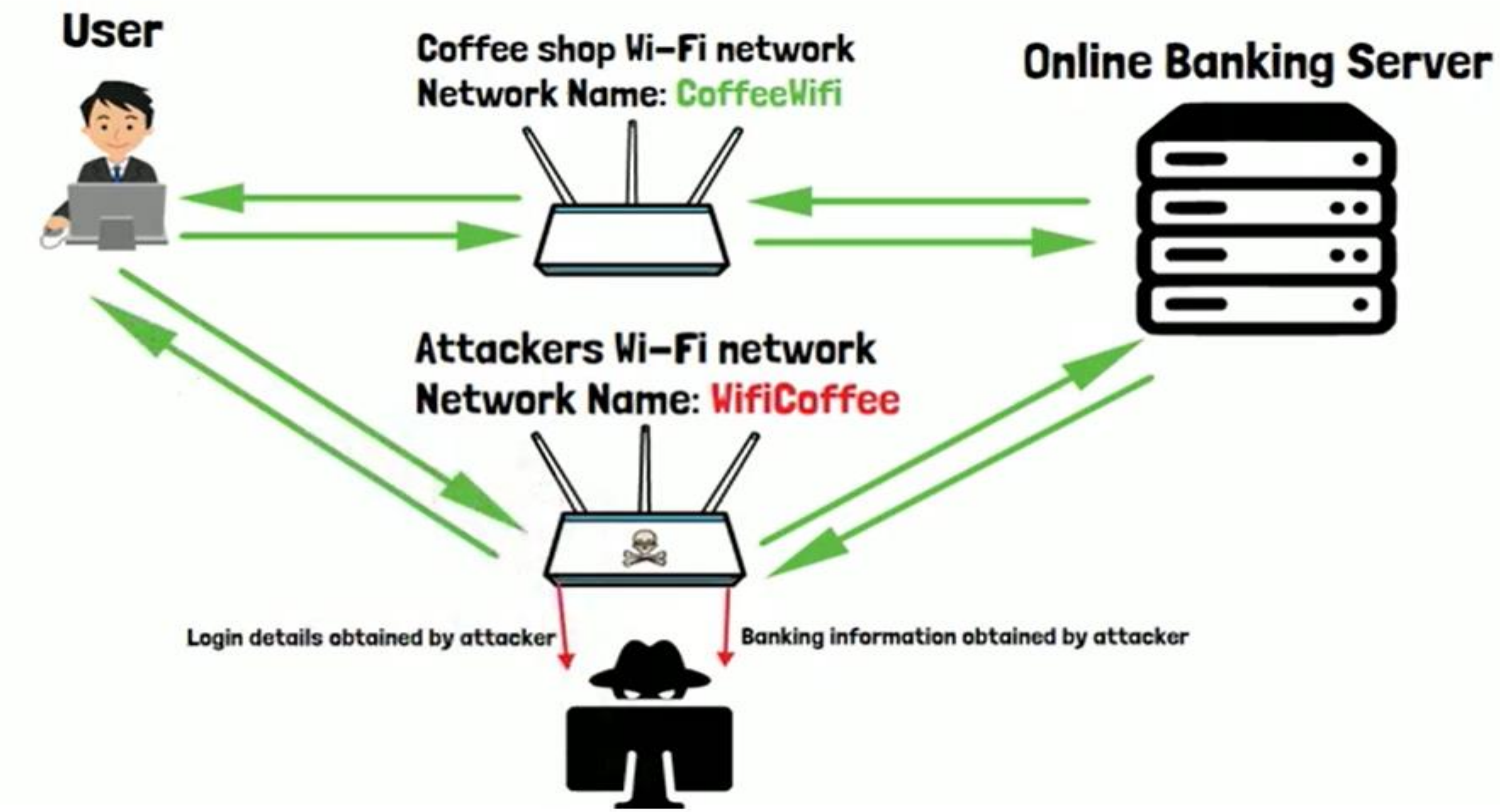
Man-In-The-Middle (MITM) Attack

Email hijacking

- **Email hijacking** is a **cyberattack** in which a cybercriminal **gains** access to a user's **email account**.

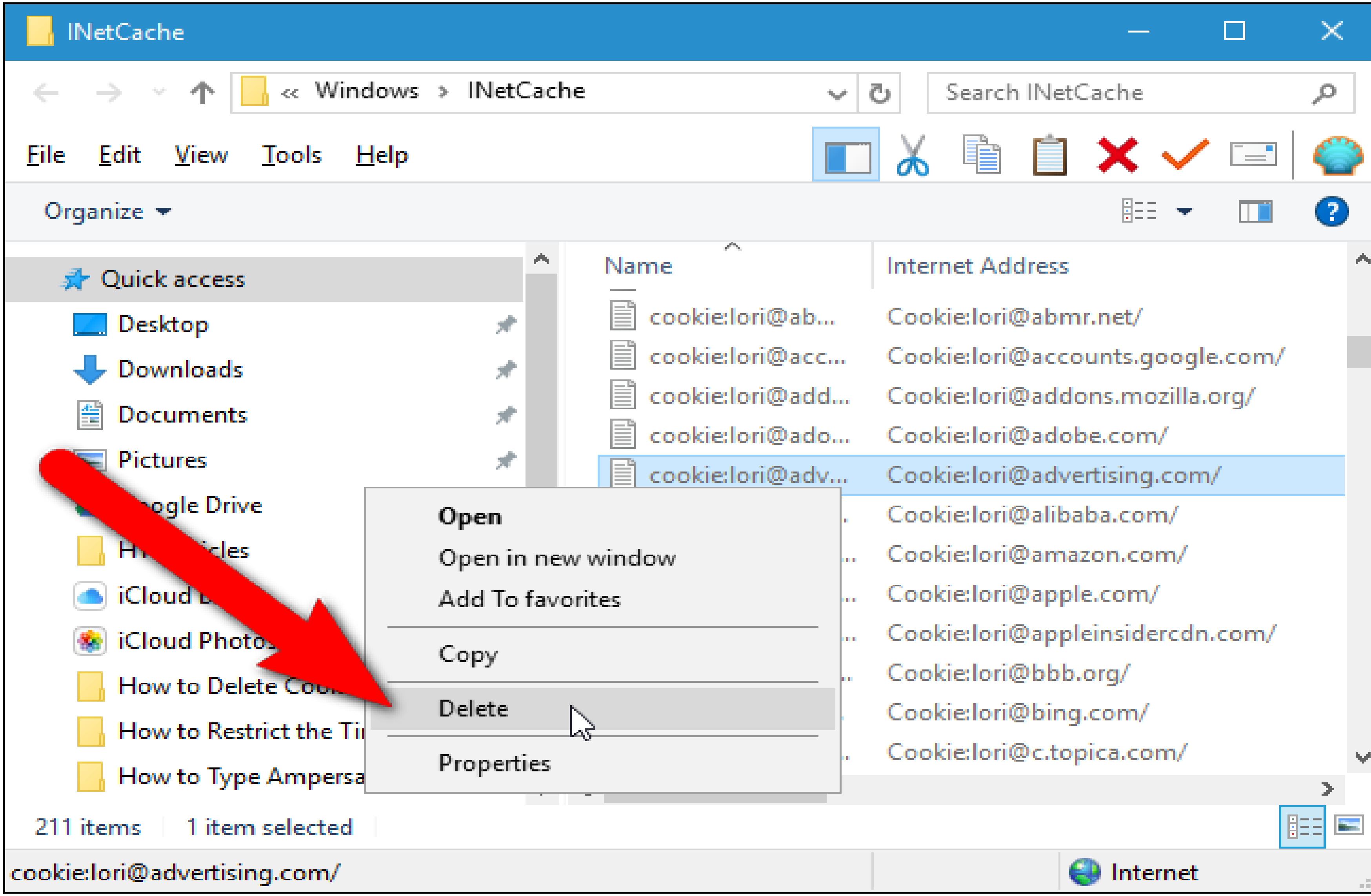
Man-In-The-Middle (MITM) Attack

Wi-Fi network eavesdropping



- **Wi-Fi Eavesdropping** can involve a hacker stealing data while on a **public** or **unsecured** Wi-Fi **network**.

Man-In-The-Middle (MITM) Attack



Stealing cookies browsers

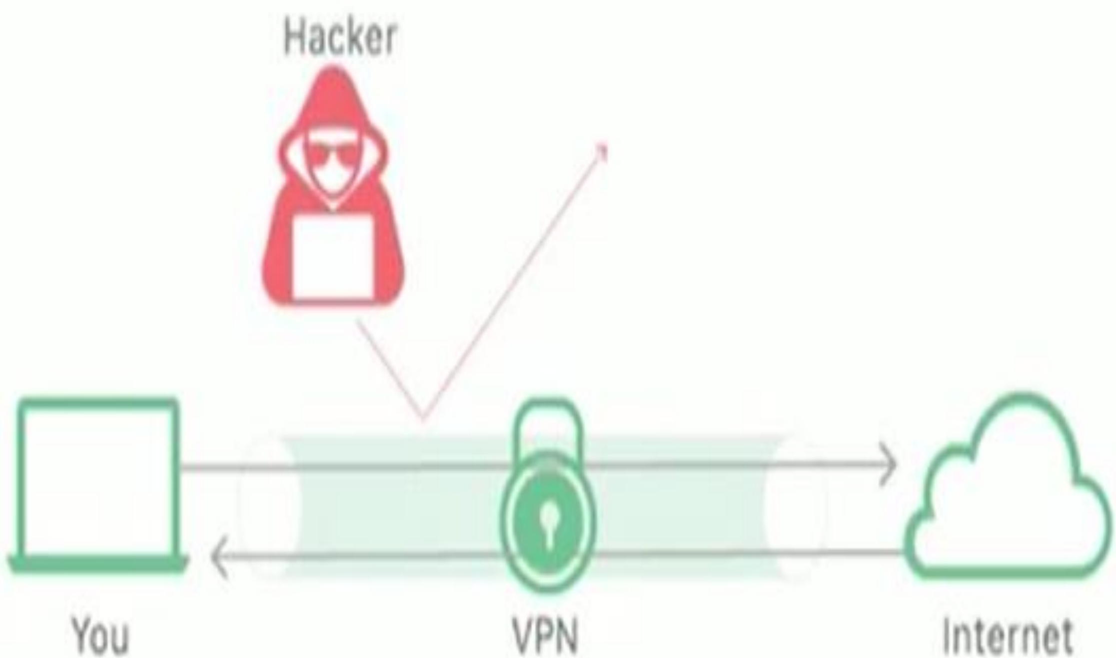
How to avoid MITM Attacks?

1	Use a spam filter.	✓
2	Don’t click emailed links or attachments.	✓
3	Use bookmarks for frequently visited pages.	✓
4	Manually search for frequently visited URLs.	✓
5	Use a trusted antivirus program.	✓

How can we protect ourselves from Wi-Fi Eavesdropping?

- Do not use public wifi for anything confidential or sensitive

- Use a VPN (Virtual Private Network)

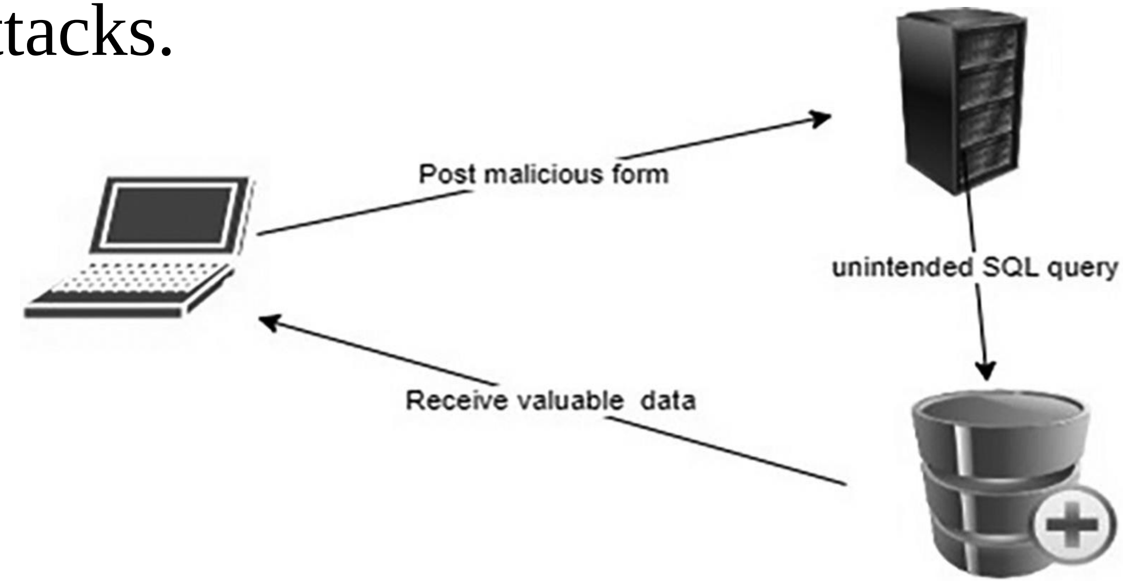


SQL Injection

- A type of **malicious practice** to **steal** the **valuable data** from the **database server**.
- This method **exploits** the **vulnerabilities** in the **traditional Active Server Page** (ASP) **websites**, **PHP applications**, and **SQL server** forms.
- The **malicious** user **appends** an **SQL** command in the **back**-**end** of the **SQL** form field.
- The **objective** is to **break** the **original** SQL **script** and run the **malicious** script **attached** with the **SQL form**.

SQL Injection

- **IT banks, e-transaction website**, and **government websites** were the top three fields badly **affected** by the **web application** attacks that **predefined** by **SQL injection** attacks.



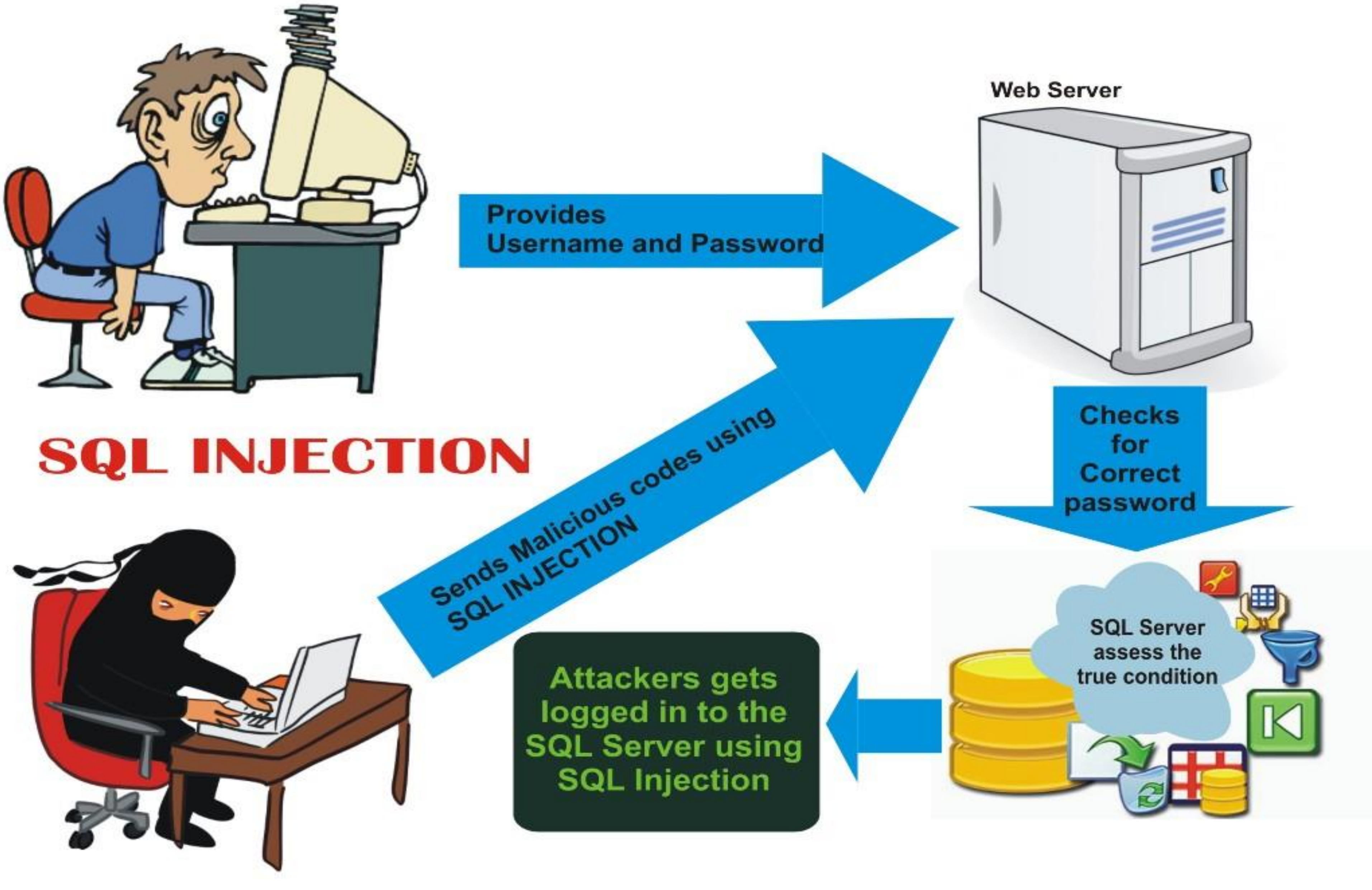
Example:

Select * from users where username = ‘user ‘;

Replaced with

Select * from users where username = ‘ ‘ OR ‘1’= ‘1’;

SQL Injection



Spamming

- **Spamming** in the IT **field** is the name of **sending junk mails** and **messages** to the **users** in **bulk without** getting **permission** from the **users**.

- The **hackers** also use **spamming** for **spreading malware**, **viruses**, **phishing**, **Trojans**, **worms**, and **spyware**.

Spamming

- **Spamming** is a **widespread** form of **malicious attacks** used to send the **unauthorized messages** through **different** modes of **messaging** such as instant **messages**, **emails**, **social network** messages, **ads**, **mobile** phone **messages**, and **social groups**.

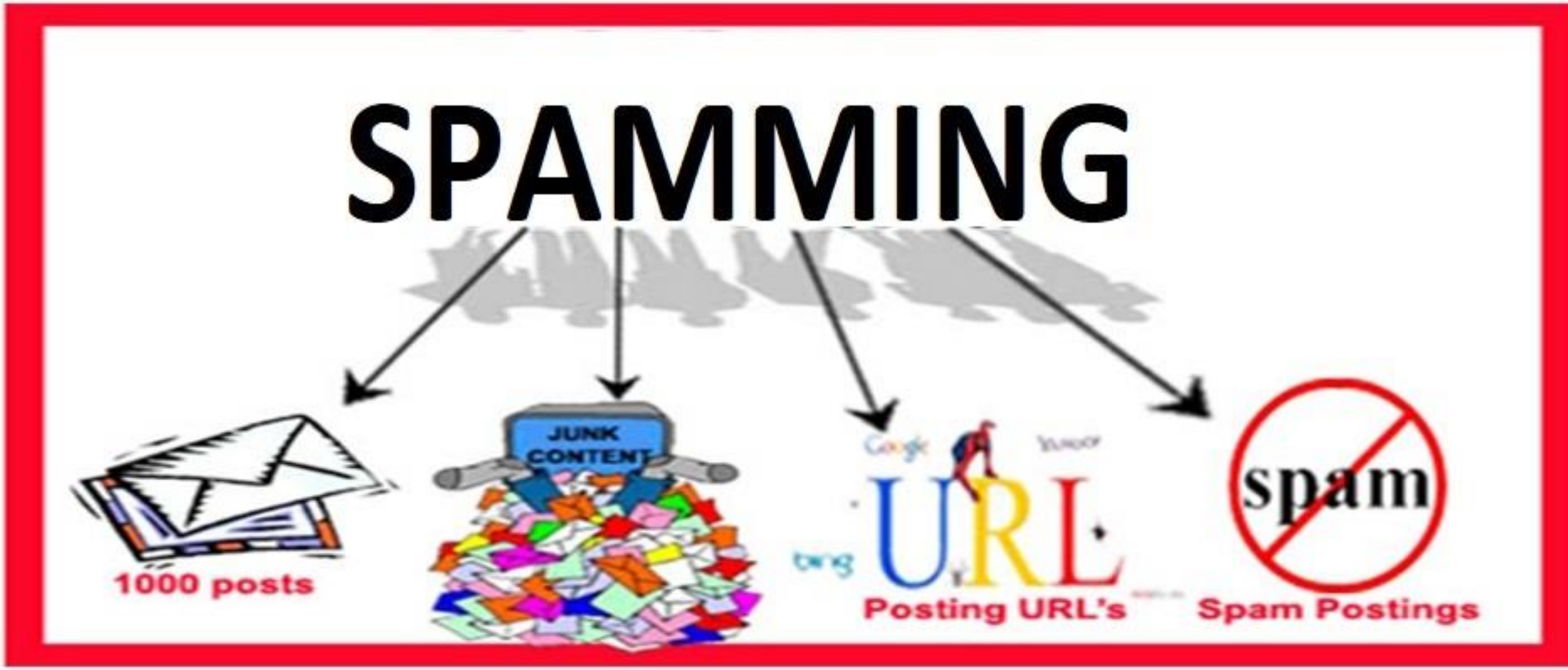
Spamming

- According to the **research**, more than **14.5 billion emails** were **sent** out on the **Internet** daily basis in the **starting months** of **2018** .

- Based on research, **in 2024,** the estimated number of emails **sent** and **received** daily was around **361.6 billion.**

- The **senders** of **spam emails** all around the **world collectively earn** about **US $7,000 per** day.

Spamming



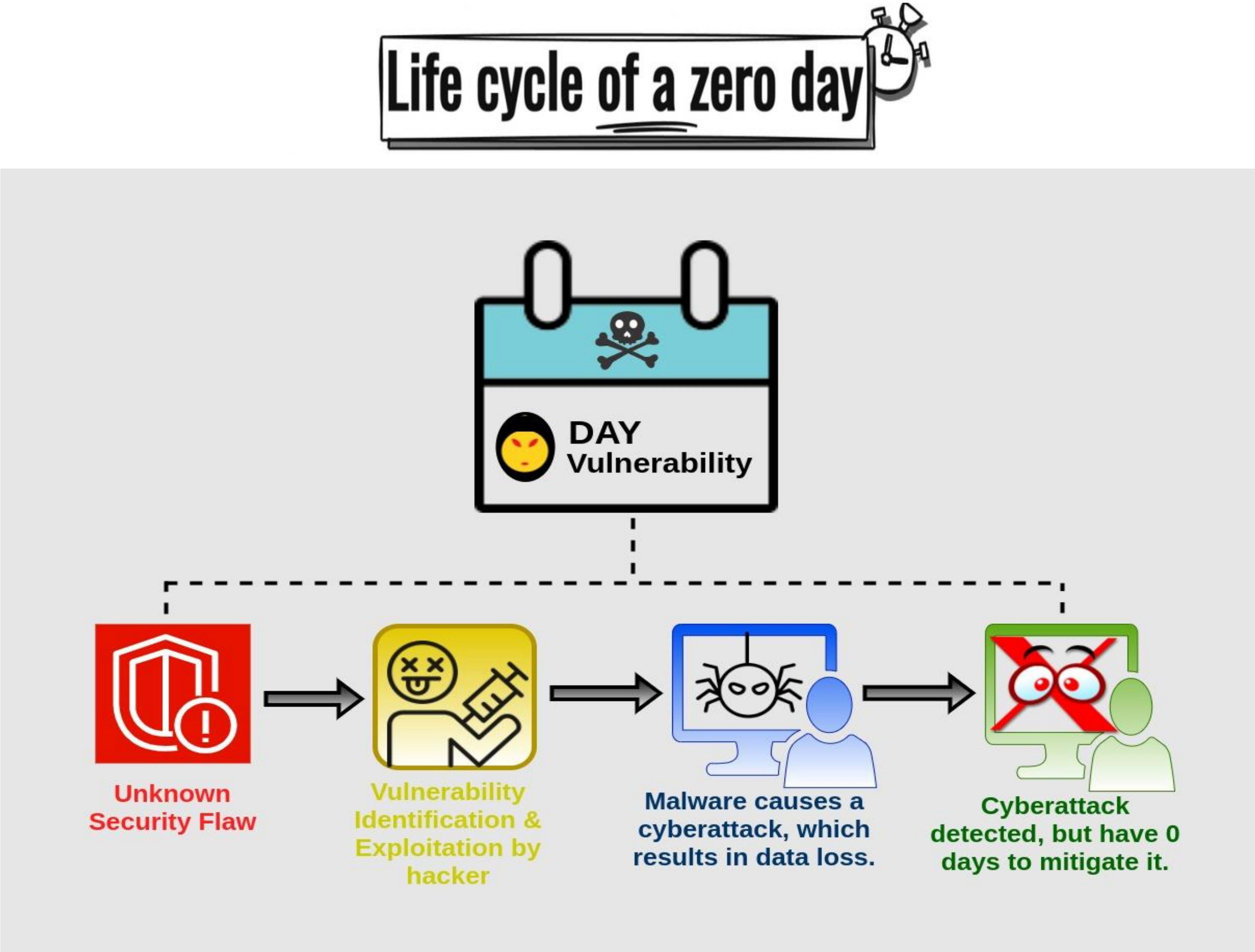
SPAMMING TOOLS

Zero-Day Exploitation

- It is a **vulnerability** in the **computer software system** that is **known** exactly on the **same day** when the **malicious attacks** exploit that **vulnerability**.

- In this attack, there is **almost no time** to patch up **(correct)** the **vulnerability** of the **software**. So, **no time** was **available** for the **software engineers** to **solve this issue.**

- In **2017 Google** has paid over **US$3 million** in the **name** of **bug bounty rewards** called **"white hat hackers."** While in 2025 paid over **US$17.1 million.**



Life cycle of a zero day
DAY
Vulnerability
Unknown Security Flaw
Vulnerability Identification & Exploitation by hacker
Malware causes a cyberattack, which results in data loss.
Cyberattack detected, but have 0 days to mitigate it.

Phishing

- It is a type of **cyberattack** in which the **targeted person** is **flooded** with **emails** **similar** to those from their **banks**, **insurance companies**, and **other service providers**.

- The hacker **targets** people through **emails** to get their **sensitive** and **personal information** related to **their financial** and **other** account **information**.

Phishing

- The main target of the **phishing** attack is to get the **information** about the **credit card number**, **ATM pin codes**, **user name**, **passwords** and the **related information**.

- Once the **information** has been **collected**, the **hackers** use that **information** to **steal** the **money** or other **valuable digital assets**.

- This **attack** is **normally** used for the **financial theft** of the **bank accounts**.

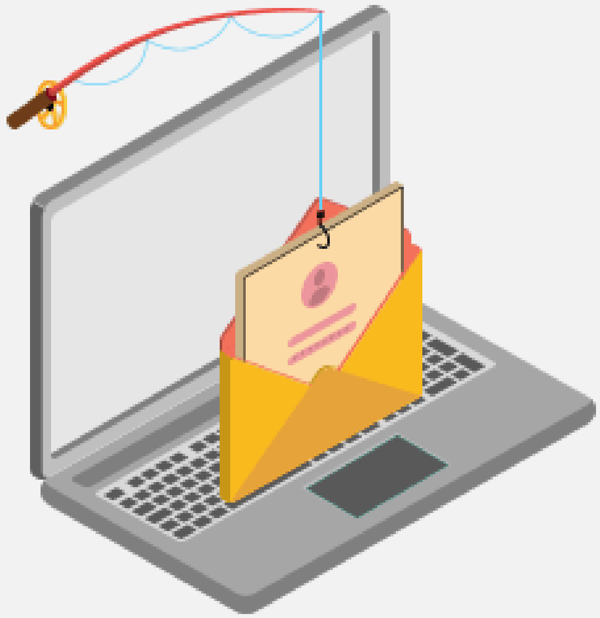
Phishing

- There are **three types** of **phishing** used in the **modern** activities as listed below:
  - **Telephone calls** commonly referred to as **voice phishing**, or vishing.
  - **Emails** referred to as **general phishing**.
  - **Small** text **messages** (SMS) referred to as **smishing**.

Phishing

Spoofing Vs. Phishing

Spamming Vs. Phishing



PHISHING

Phishing attacks are designed to steal user private information such as passwords, bank accounts, credit card details, etc.

SPAM

Spam is unsolicited email or junk having irrelevant content or commercial advertising contents.

Cyberattacks Mind map

Cyberattack	Confidentiality (C)	Integrity (I)	Availability (A)
MITM	Yes	Yes	No
Phishing	Yes	Sometimes	No
Spamming	No	No	Yes
DoS / DDoS	No	No	Yes
SQL Injection	Yes	Yes	Yes
Zero-Day Exploitation	Yes	Yes	Yes

Questions & Answers

Q1.What is DoS Attack? What are the general symptoms?

A1: The hacker actively exploits the **server vulnerability** and sends a lot of **automated requests** and **messages** to that **particular server** to **respond**. So, The **server** gets **choked** and **stops working normal**.

The major symptoms of being the victim of DoS attacks (for a legitimate user) include the following:

- Inability in accessing a website.
- Delay in accessing online service.
- Huge delays in file opening on the websites.
- Increased volume of spam emails.

Questions & Answers

Q2. What is MITM?

A2: In the “Man-in-the-Middle” or **MITM Cyberattack**, the hacker **intercepts the normal connection** between the user and the web server without any knowledge of both user and server.

- **The legitimate communication** link between the two entities is **exploited**, **intercepted**, and **decrypted** to **steal** the **personal information** for **malicious use.**

Task (15 Min)

- Find the **plaintext** by using the **Rail Fence** cipher to **decrypt** the following:

Ciphertext: HEEMIRTGIIADCMANEGEEESSDESSI

Key: 5

thanks for
listening!